

MATH 3670 Homework 2

Saahas Swaroop

September 12, 2025

24

$$P(G|G) = \frac{P(GG)}{P(G)} = \frac{.25}{.5} = .5$$

25

$$P(F) = .52, P(C) = .05, P(FC) = .02.$$

a

$$P(F|C) = \frac{P(FC)}{P(C)} = .4$$

b

$$P(C|F) = \frac{P(FC)}{P(F)} = \frac{1}{26}$$

27

$$P(D) = \frac{.05 + .01}{2} = .03$$

$$P(D|D) = \frac{P(DD)}{P(D)} = \frac{.03 \cdot .03}{.03} = .03$$

29

a

$$P(A) = P(A|W)P(W) + P(A|W^c)P(W^c)$$

$$P(A) = .85 \cdot .9 + .2 \cdot .1 = .785$$

b

$$P(W^c|A^c) = \frac{P(A^c|W^c)P(W^c)}{P(A^c)}$$

$$P(W^c|A^c) = \frac{.8 \cdot .1}{.215} = .372$$

31

$$P(D|V_R) = \frac{P(V_R|D)P(D)}{P(V_R)} = \frac{.125 \cdot .4}{.59} = 0.085$$

35

$$P(A) = P(A|G)P(G) + P(A|Avg)P(Avg) + P(A|B)P(B) = .05 \cdot .2 + .15 \cdot .5 + .3 \cdot .3 = 0.175$$

$$P(G|A^c) = \frac{P(A^c|G)P(G)}{P(A^c)} = 0.2303$$

$$P(A|A^c) = \frac{P(A^c|A)P(A)}{P(A^c)} = 0.5152$$

39

a

$$P(H)^3 + P(T)^3 = .25$$

b

$$P(A \cup B) = P(A) + P(B) - P(A \cap B) = .25 + .25 - .0625 = 0.4375$$

41

$$P(S) = .5^n$$
$$P(C|S) = \frac{P(S|C)P(C)}{P(S)} = \frac{1 \cdot .5}{.5^n} = .5^{1-n}$$